

Midnight Buildathon · Wave 1

# One pledge per invoice. No lender sees another's book.

A registry on Midnight that refuses to let the same receivable be financed twice, while each lender sees only its own deals.

## The problem

# The same invoice, pledged to two lenders. Both pay out.

### Lenders can't compare notes

Catching a double pledge means pooling books. A lender's clients and pricing are its business, so nobody shares.

### Hash registry: bypassed

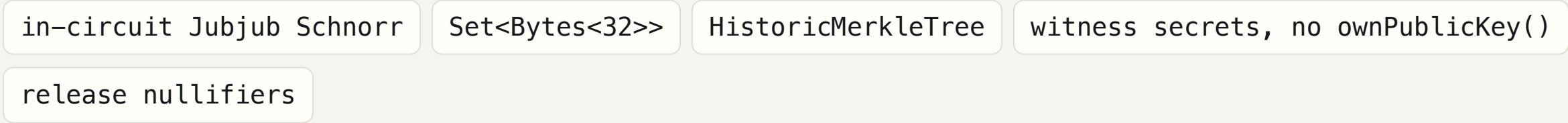
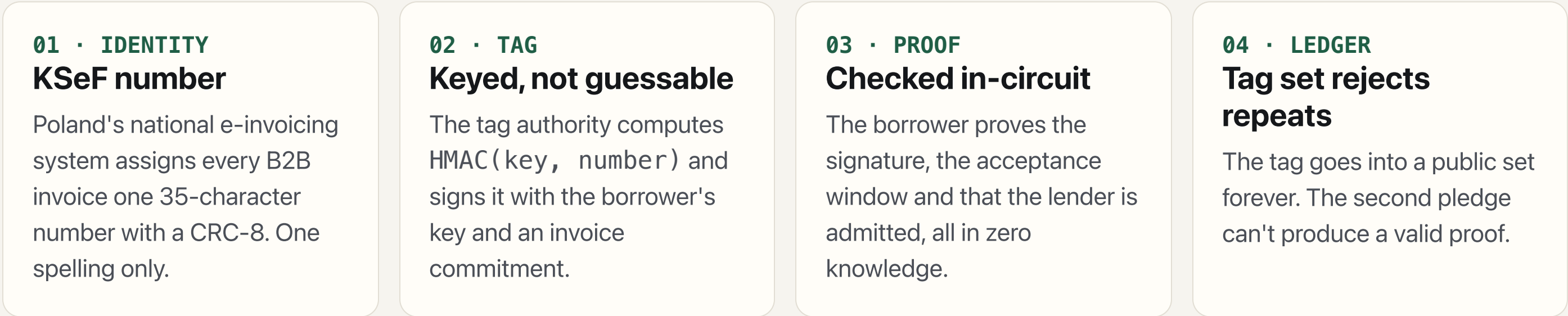
FV/2026/09/0412 and FV-2026-09-0412 hash differently. Each bank's spelling is a "new" invoice.

### Hash registry: leaks

Anyone holding an invoice hashes it and learns from the public chain whether, and when, it was financed.

How it works

# A canonical ID, a keyed tag, and a proof checked on Midnight.



Selective disclosure

# Everyone learns exactly what they need.

| PARTY            | LEARNS                                                    | NEVER LEARNS                             |
|------------------|-----------------------------------------------------------|------------------------------------------|
| Financing lender | <b>full invoice</b> checked against the signed commitment | Other lenders' deals                     |
| Rival lender     | <b>"already pledged"</b>                                  | By whom, how much, which terms           |
| Public chain     | Opaque tag, note commitment, counts                       | <b>invoice, amount, borrower, lender</b> |
| Tag authority    | Which invoices it attested                                | Which lender financed them               |

Pinned by privacy-leak tests that search the full public state and every circuit transcript for each private value.

# A working registry, tested like it handles money.

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circuits compiled with proving keys

95

tests: adversarial witnesses, forgery,  
privacy leaks

6

Preprod transactions: deploy, 2 lenders, 2  
pledges, release; the double pledge  
rejected before proving

## Proven

- Five double-pledge variants rejected
- Ten forgery attempts rejected, including a malicious prover
- Stolen attestations can't be used by anyone else
- Live registry page reads the Preprod contract, no wallet

## Honest limits

- Single tag authority (threshold in Wave 3)
- Synthetic KSeF numbers until the test-environment adapter
- Facilities, draws and Lace UI are Wave 2

# Start where every invoice already has one ID.

## Beachhead: factoring on KSeF

Mandatory national e-invoicing gives every Polish B2B invoice a canonical number. Buyers: factors, supply-chain-finance banks, private-credit funds. Distribution: e-invoicing access points that already connect everyone.

Useful with just two lenders. No consortium needed.

## Next

- **Wave 2:** Lace-connected borrower desk and lender inbox on Preprod, facilities and borrowing-base draws, encrypted invoice delivery, KSeF test-environment adapter, practitioner interviews
- **Wave 3:** threshold tags (no single party can tag), pledge assignment, reusable uniqueness module for any guessable-ID registry